Name: Bharath Arun Gandhimani **Student ID:** 35501308
Applied Session: 7 | Teaching Associates: Michael Niemann & Ayden Zhou

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Project Title: From Roads to Classrooms: Educational Infrastructure Investment, Socioeconomic Equity, and Household Travel Behavior in Victoria (2023–2025)

Aims:

This visualisation communicates three connected findings to Victorian policymakers and transport planners:

- Education infrastructure investment broadly tracks where households are located across Victoria's LGAs
- Higher investment tiers correlate with greater car dependency, not active or public transport uptake
- Neighbourhood disadvantage — not investment level alone — is the stronger predictor of how households travel

The goal is to show where government spending and community transport need are aligned, and where they are not.

Motivation:

Victoria spends billions annually on regional infrastructure — but investment alone does not guarantee equity. If funding concentrates in already well-connected areas, or if economically disadvantaged households cannot access viable transport options, the gap between government intention and community outcome widens. This project combines VISTA 2024–25 travel survey data, the 2023–24 State Budget Infrastructure Investment Map, and ABS SEIFA 2021 indexes to assess whether infrastructure decisions are spatially and socially aligned with real household travel demand — and what that means for equity in daily mobility across Victoria.

SHEET 1 — IDEAS

Bharath Arun Gandhimani | FIT5147 DVP | 10 / 05 / 2026 (revised 02 / 06 / 2026) **FDS**

IDEAS

FILTER

- **#9** → Heatmap: poor at differentiating small % change across cells.
- **#10** → White-cell table: individual unit encoding becomes unreadable.
- **#2** → Toggle: interaction complexity without clear payoff.
- **#17** → Static log page: needs interactivity to be meaningful.
- **#6** → Lollipop: redundant with bar + dot plot combination.
- **#8** → Diverging bar: realistic neutral midpoint may not exist in this data.
- **#13** → Scroll narrative: JavaScript dependency — excluded per course constraints.
- **#23** → Linked highlight: patterns matter more than individual row comparison.

CATEGORISE

Ideas that Set Context
`#24, #25, #12`
Orients the reader before analysis. Establishes scale, frames questions, shows what is most prominent.

Ideas that Frame the Narrative
`#25, #26`
Tells the reader what data can and cannot show. #26 serves double duty — contextualises entry point and frames the whole narrative.

Ideas that Show Comparisons
`#1, #2, #3, #5`
Lets the reader see differences between groups or across an ordered axis.

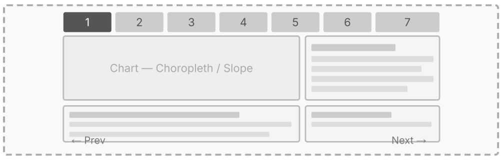
Ideas that Show Distribution & Variability
`#4, #6`
Reveals spread and shape behind summary stats. #11 can serve as companion across subgroups and as a layout device.

Ideas that Give Reader Control
`#18, #19, #20, #21`
Interactive controls. Augment charts, not standalone. Risk: complexity without payoff for a policymaker audience.

COMBINE & REFINE

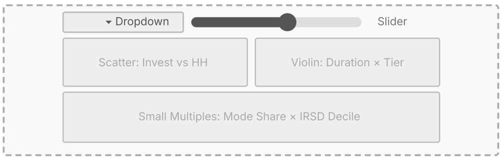
Design A — Sequential Stepper

Fully author-driven. Fixed forward-only path. Each step: one chart (choropleth / lollipop / slope / grouped bar / scatter) + narrative text panel. Reader cannot skip ahead.



Design B — Question-Answer Dashboard

Fully reader-driven. Three panels simultaneously visible using distinct encodings: scatter (bivariate), violin (distribution), small multiples (faceted comparison). Scoped filters per panel.



Design C — Martini Glass

Guided scroll narrative (author-driven) through three headline-and-chart panels, then a state-change button unlocks an LGA Explorer. Author controls the evidence sequence; reader controls the depth. Chart family: Cleveland dot plot (ranked LGAs), slope/connected line chart (mode share across tiers), connected dot plot (car vs PT time penalty). Note: Sheet 1 originally listed a bump chart for [P2]; this was revised to a slope/connected line chart in Sheets 4–5 for legibility across four discrete ordered tiers. Revision documented in Sheet 5 Deviations.

QUESTIONS

Q1
Does any design give a policymaker a clear sense of what action to take?
Design A ends at a clear policy implication. B and C leave the reader to draw their own conclusions, risking unresolved questions for a non-expert audience.

Q2
How much narrative text does a policymaker need alongside each chart?
Design A allocates explicit text panels per chart. B and C prioritise chart real estate over narration — risky for a policymaker who needs framing to act.

Q3
If a reader enters mid-way, can they understand without reading earlier sections?
Design B handles this best — each panel is self-contained. Design A breaks without sequential reading. C partially handles it via the persistent headline numbers in the title panel.

Q4
Does Design A's fixed sequence risk losing a reader interested in only one question?
Yes, but acceptable — policymakers need the full picture to act. Author-controlled narrative is appropriate for this audience.

Q5
Does Design B's simultaneous visibility create cognitive overload?
Yes. Showing all three questions at once without a guided entry overwhelms a non-expert reader. Works for analysts, not policymakers.

Q6
Does Design C's Martini Glass structure balance author-controlled narrative with reader-driven exploration, and is the two-state transition clear enough to prevent premature exploration?
Mostly yes. The narrative panels deliver the complete argument before the exploration zone unlocks, so readers who exit early still receive the full message. The main risk is that a persistent right column signals exploration before it is earned — the overlay copy and [D] button must be calibrated to preserve the author's intended reading order.

Q7
How do we communicate the structural data gaps without it looking like incomplete analysis?
Frame the 29-LGA coverage gap as a policy-relevant finding: VISTA does not survey regional Victoria, meaning policymakers currently lack travel behaviour evidence to evaluate investment in those areas. Present as a prominent callout, not a limitation footnote — the gap itself is part of the story.

DVP PART 1 FEEDBACK — Revision Record

Michael Niemann identified that the three alternative designs were too similar — primarily layout variants rather than genuinely distinct design approaches. What structural changes were made for DVP Part 2?

The revised designs differ in three dimensions simultaneously. (1) **Narrative genre:** Design A is fully author-locked with forward-only stepper navigation; Design B is fully reader-controlled with simultaneous scoped panels; Design C uses a two-state Martini Glass — author-controlled in State A, reader-controlled in State B. (2) **Chart family:** Design A uses geographic and distribution encodings (choropleth, lollipop, slope, grouped bar, scatter); Design B uses bivariate, distributional, and faceted encodings (scatter, violin, small multiples); Design C uses ranking, trend, and paired-comparison encodings (Cleveland dot plot, slope/connected line chart, connected dot plot). No chart type appears in more than one design. Note: Design C originally showed a bump chart for [P2] at brainstorming stage; this was revised to a slope/connected line chart during realisation (Sheet 5) for legibility across four discrete ordered tiers — the change is documented in Sheet 5 Deviations. (3) **Interaction model:** Design A has no filters — locked forward navigation only; Design B has asymmetrically scoped cross-panel filters; Design C has no filters — only a state-change toggle between guided narrative and LGA personalisation. These differences operate at the level of spatial grammar, perceptual task, and interaction philosophy.

SHEET 2 — DESIGN A: SEQUENTIAL STEPPER

Bharath Arun Gandhimani | FIT5147 DVP | 12 / 05 / 2026 (revised 02 / 06 / 2026)

FDS

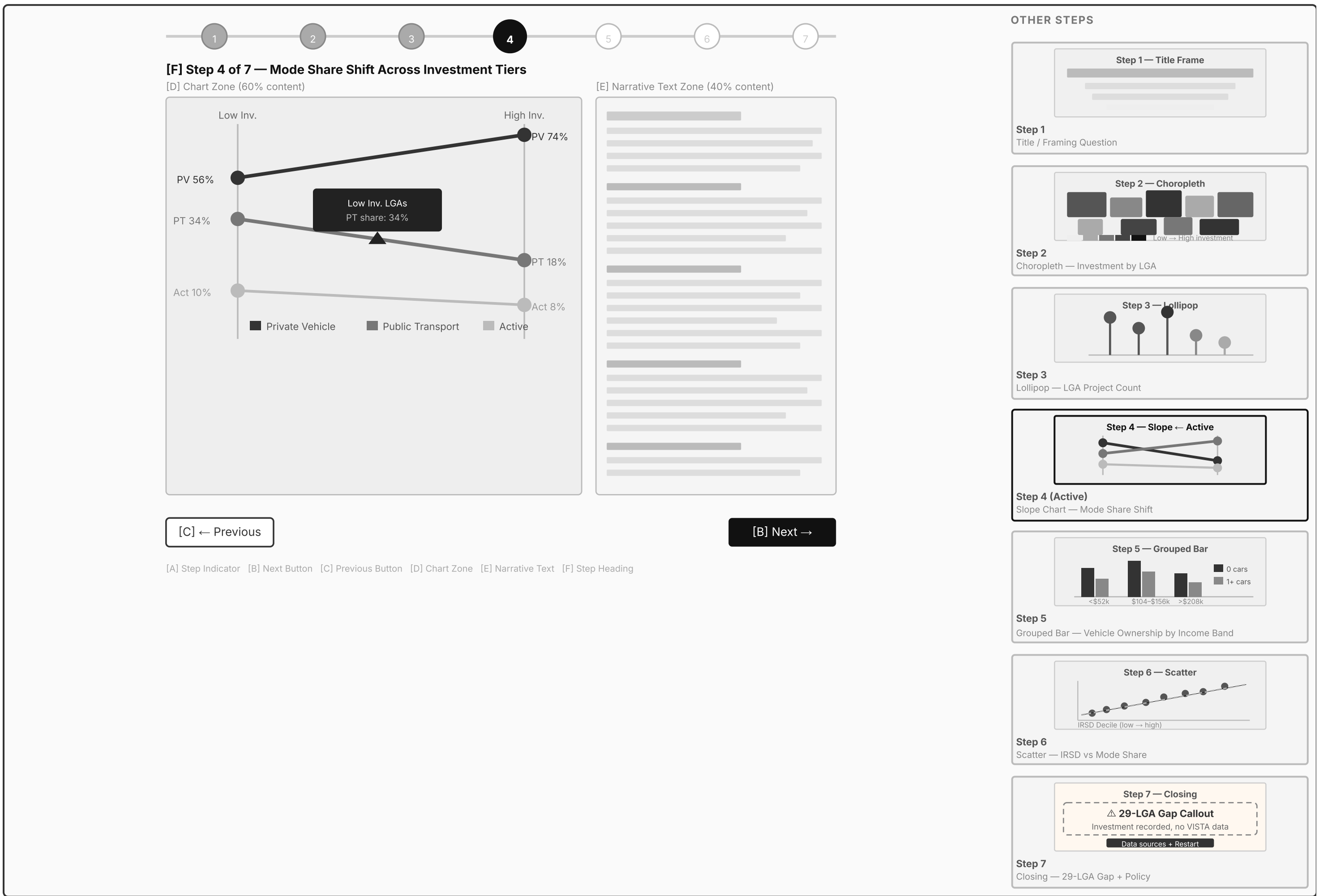
INFORMATION

Title: Design A — Sequential Stepper | **Sheet No:** Sheet #2
Author: Bharath Arun Gandhimani | **Date:** 12/05/2026
Description: Fully author-driven interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity, and household travel behaviour in Victoria. Targeted at policymakers and transport planners. Reader progresses through a fixed 7-step sequence — one analytical layer per step.

DESIGN RATIONALE

The stepper enforces a deliberate reading order — investment geography is established first before socioeconomic or behavioural patterns are introduced. This prevents a policymaker from jumping to conclusions without the geographic context that frames the entire argument. Chart types (choropleth, slope chart, lollipop) are drawn from Sheet 1 Ideas #7, #6, and the geographic/map category respectively. No filters or dropdowns — narrative control stays with the author. Step 5 shows a grouped bar chart of vehicle ownership by household income band (under $52k, $104–$156k, over $208k), with 0-car and 1+ car households grouped side by side — this variable is directly available in hh_seifa_final.csv. Chart family (choropleth, lollipop, slope, grouped bar, scatter) is entirely distinct from Design B (scatter, violin, small multiples) and Design C (Cleveland dot plot, slope/connected line, connected dot) — these three designs use non-overlapping visual encodings, satisfying the FDS requirement for genuine chart-level distinctness across Sheets 2–4.

LAYOUT



OPERATIONS

[B] Click "Next →"
⇒
Content in [D] and [E] fades out. Next step's chart and narrative text fade in. Step indicator [A] advances — current circle fills solid grey (visited), next circle activates black. On the final step the Next button is replaced by a "Restart" option.

[C] Click "← Previous"
⇒
Returns to the prior step. [A] updates to reflect the previous active step. [C] is hidden when the reader is on Step 1 — there is no "before" the opening frame.

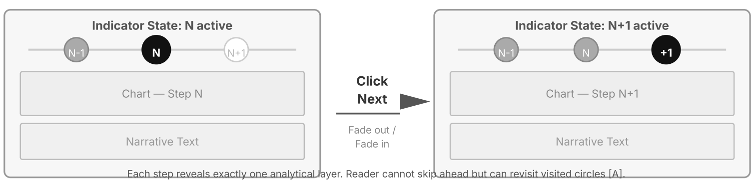
[A] Click a visited step circle
⇒
Jumps directly to that completed step for review. Unvisited future circles remain greyed out and unclickable — the reader cannot skip ahead to content they have not yet reached.

[D] Hover on chart element
⇒
A tooltip appears showing exact values — percentage, count, LGA name. This is the only interactivity within a step. No filters, no dropdowns, no view changes are available.

[A] Step circle — state logic
⇒
Visited circles show filled grey. Active circle shows filled black. Unvisited future circles show white with light grey border and are unclickable. State maintained for the full session.

Step 7 — Final step reached
⇒
The 29-LGA gap callout and policy implication text are revealed. Data source links and a "Restart" button appear. The [B] Next button is hidden. Reader has completed the full narrative arc.

FOCUS / PARTI — STEP TRANSITION MECHANISM



The closing step only makes sense once all prior evidence has been absorbed — this is what separates Design A from B and C. Chart types per step: Step 2 → Choropleth (geographic category, Sheet 1); Step 3 → Lollipop (Sheet 1 Idea #6); Step 4 → Slope Chart (Sheet 1 Idea #7).

Step Circle States
- ● Visited — filled grey
- ● Active — filled black
- ○ Locked — white, unclickable

Label Reference
[A] Step Indicator
[B] Next Button
[C] Previous Button
[D] Chart Zone
[E] Narrative Text
[F] Step Heading

DISCUSSION

Pros

- Sees evidence in exact order as intended — no confusion about where to start.
- One chart, one explanation per step keeps things simple and focused for a non-technical policymaker audience.
- Opens with geographic framing (choropleth). Closes with what remains unknown (29-LGA gap). Clean narrative arc.
- Works well with non-technical audiences like policymakers who need framing before data.
- Easy to implement in R Shiny using a reactive step counter with conditional panel rendering.
- Chart types in this design — choropleth, lollipop, slope, grouped bar, scatter — address geographic and categorical comparisons. This is a different perceptual task family from Design B (distributional / bivariate) and Design C (ranking / trend / paired comparison), ensuring genuine design-space breadth across Sheets 2–4 as required by the FDS methodology.
- Hover tooltips as the sole interactivity keeps implementation scope controlled and avoids cognitive overload.

Cons

- Cannot compare two charts side by side — reader must go back and forth.

SHEET 3 — DESIGN B: QUESTION-ANSWER DASHBOARD

Bharath Arun Gandhimani | FIT5147 DVP | 13 / 05 / 2026 (revised 02 / 06 / 2026) **FDS**

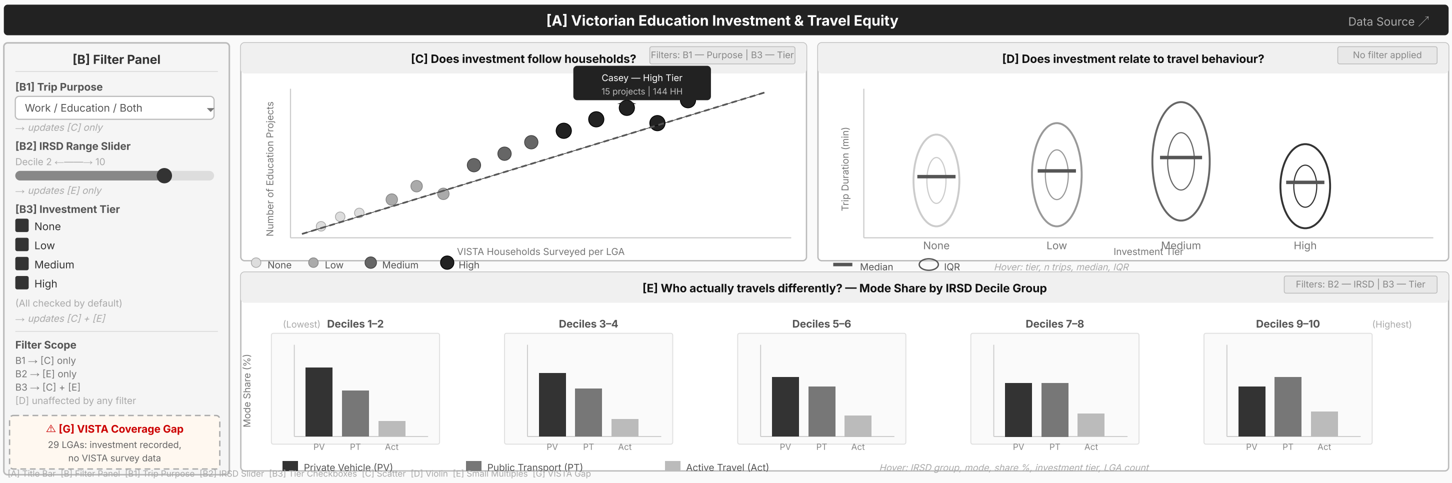
INFORMATION

Title: Design B — Question-Answer Dashboard | **Sheet No:** Sheet #3
Author: Bharath Arun Gandhimani | **Date:** 13/05/2026
Description: Fully reader-driven dashboard communicating findings about educational infrastructure investment, socioeconomic equity, and household travel behaviour in Victoria to policymakers. Three chart panels are visible simultaneously — each poses an explicit question and answers it with a distinct encoding. A left-side filter panel scopes each filter to specific panels only.

DESIGN RATIONALE

Each panel uses a different encoding matched to its analytical task: scatter plot (bivariate investment-vs-household relationship), violin plots (distribution shape across investment tiers), and small multiples (faceted mode share by IRSD decile). Chart types drawn from Sheet 1 Ideas #5 (scatter), #4 (violin), #11 (small multiples). Filters are intentionally scoped — [B1] affects [C] only, [B2] affects [E] only, [B3] affects [C] and [E] — to prevent unintended cross-panel interference. [D] is deliberately unfiltered so the reader always has a stable distributional reference. No chart type in this design (scatter, violin, small multiples) appears in Design A or Design C — the three designs collectively cover geographic, distributional, bivariate, faceted, strip, ranking, and paired-comparison encodings across Sheets 2–4.

LAYOUT



OPERATIONS

[B1] Select Trip Purpose dropdown
⇒
[C] scatter updates to trips matching selected purpose (Work / Education / Both). [D] and [E] unchanged — purpose-level breakdown not available at LGA-level for violin or faceted mode share.

[B2] Drag IRSD Range Slider
⇒
[E] small multiples grey out facets outside selected decile range — highlights relevant band without removing context. [C] and [D] unchanged.

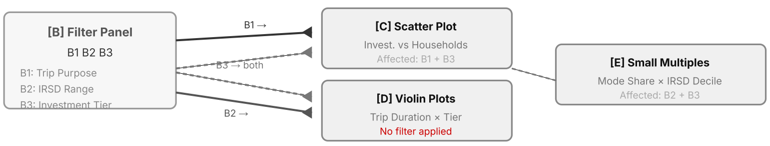
[B3] Uncheck Investment Tier
⇒
[C] removes points for unchecked tier and re-draws trend line. [E] removes bars for that tier within each facet. [D] unaffected — violin shows full distribution regardless of tier.

Hover on point in [C] scatter
⇒
Tooltip: LGA name, tier, project count, household count, trip purpose. Hovered point highlighted; all others dim to 30% opacity.

Hover on violin body in [D]
⇒
Tooltip: tier label, median duration (min), IQR range, n trips. Median and IQR markers always visible statically — hover supplements, does not replace.

Hover on bar in [E] small multiples
⇒
Tooltip: IRSD decile group, mode type, share (%), n LGAs in facet. Same mode bar highlights across all visible facets simultaneously for cross-facet comparison.

FOCUS / PARTI — FILTER COORDINATION FLOW



Parti note

3 questions → 3 encodings
Scatter: bivariate relationship
Violin: distribution shape
Small mult: faceted comparison

[D] unfiltered — stable distributional reference
[E] not affected by B1

Three questions, three completely different encodings — no chart type is reused across panels. Filter scope is asymmetric by design: each filter targets only the panel where it is analytically meaningful. [D] is permanently unfiltered to give readers a stable distributional baseline regardless of how they interact with [B1] or [B3].

Panel Summary
[C] Scatter: X = VISTA households surveyed, Y = Education project count, Colour = Tier
[D] Violin: X = Tier (4 groups), Y = Trip duration (min), Overlay = median+IQR
[E] Sm. Multiples: Facet = IRSD decile (5), X = Mode, Y = Share (%)

Label Reference
[A] Title Bar [B] Filter Panel
[B1] Trip Purpose Dropdown
[B2] IRSD Range Slider
[B3] Investment Tier Checkboxes
[C] Scatter Plot Panel
[D] Violin Plots Panel
[E] Small Multiples Panel
[G] VISTA Coverage Gap Callout

DISCUSSION

Pros

- A policymaker can navigate directly to the question most relevant to their brief — no compulsory sequence to sit through before reaching the panel they care about.
- Each panel uses a distinct encoding matched to its analytical task: scatter for bivariate relationship, violin for distribution shape, small multiples for faceted comparison. No chart type is repeated across panels — each question gets its own visual language.
- Asymmetric filter scoping prevents cross-panel noise — each filter only acts where it is analytically valid, reducing reader confusion from unexpected changes.
- [D] violin plots are permanently unfiltered, giving the reader a stable distributional reference that does not change as they experiment with [B1] or [B3] — useful as an anchor point.
- Small multiples in [E] let the IRSD decile gradient emerge visually across all five facets simultaneously without any interaction needed — the pattern is visible at a glance.
- R Shiny implementation is straightforward: three independent renderPlotly() outputs, each reactive only to its scoped inputs, avoiding unnecessary re-renders.

Cons

- All three panels are visible at once with no guided entry point — a policymaker unfamiliar with the topic may not know where to start and may engage only one panel, missing the full argument.
- There is no narrative thread connecting the three questions. A reader who engages only with [C] or [D] misses the socioeconomic equity dimension that [E] communicates.
- The VISTA gap callout [G] sits at the bottom of the filter panel and is easy to overlook for a reader whose attention is drawn immediately to the larger chart panels on the right.
- Asymmetric scoping (B1 → [C] only, B2 → [E] only, B3 → [C] + [E]) requires orientation time — a first-time reader may apply a filter and be confused when not all panels update.
- [C] and [E] both involve investment tier as a variable but from different angles — without accompanying explanatory text, this could be perceived as redundant or contradictory rather than complementary.
- Unlike Design A (Sheet 2), there is no closing policy implication step — the reader must draw their own conclusions from the three panels, which may be unreliable for a non-expert audience.
- [D] violin plots show trip duration across investment tiers, but the Kruskal-Wallis test performed in the Data Exploration Project found no statistically significant differences in work-trip duration across tiers — making this panel analytically weak as a centrepiece. This was a key factor in rejecting Design B in favour of Design C: a null result does not communicate a clear policy message.

SHEET 4 — DESIGN C: MARTINI GLASS

Bharath Arun Gandhimani | FIT5147 DVP | 13 / 05 / 2026 (revised 02 / 06 / 2026) **FDS**

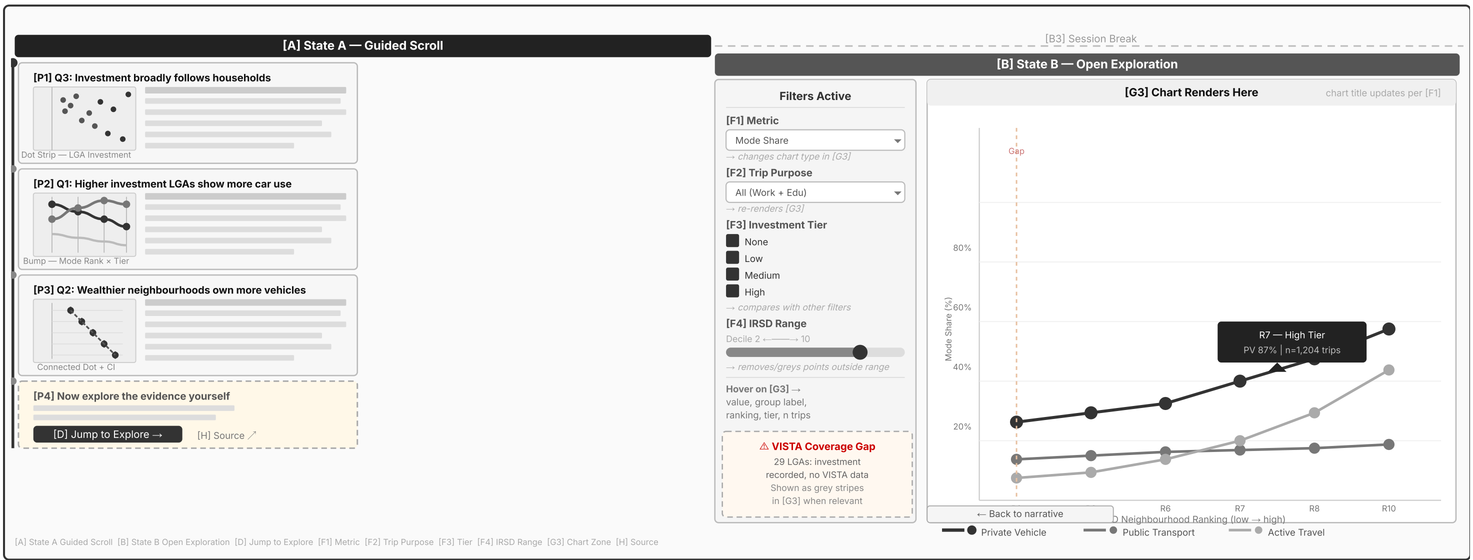
INFORMATION

Title: Design C — Martini Glass | **Sheet No:** Sheet #4
Author: Bharath Arun Gandhimani | **Date:** 13/05/2026
Description: Interactive narrative visualisation communicating findings about educational infrastructure investment, socioeconomic equity and household travel behaviour in Victoria to policymakers and transport planners. A two-state structure: State A delivers a guided scroll through four sequential narrative panels [P1–P4], each pairing a bold headline claim with a small embedded chart. A [D] button transitions the reader to State B, where filters [F1–F4] drive a live chart zone [G3] for open exploration.

DESIGN RATIONALE

The martini glass structure gives policymakers the narrative framing they need before exploration — author controls the story entry, reader controls the exit. Chart types are chosen to be fully distinct from Sheets 2 and 3 in both chart family AND analytical task — geographic/categorical (Sheet 2) vs distributional/bivariate (Sheet 3) vs ranking/trend/paired (Sheet 4): **dot strip plot** (investment dispersion across LGAs, Q3), **bump/slope chart** (mode share rank shift across investment tiers, Q1 — redesigned as a slope/connected line chart in Sheet 5 for legibility across four discrete tiers), and **connected dot plot with 95% CI error bars** (vehicle ownership by income band, Q2). [F1] Metric dropdown switches the chart type in [G3] — a mode share metric renders a connected dot plot, a duration metric renders a violin+box plot, a vehicle ownership metric renders a connected dot plot with CI bars.

LAYOUT



OPERATIONS

[A] Scroll / natural read through [P1–P3]
⇒
Reader moves through three authored panels in sequence. Each panel is a fixed layout: chart thumbnail (left) + headline claim + narrative text (right). No interaction required — author controls the evidence order. [P4] bridge panel appears at the bottom of the scroll.

[D] Click "Jump to Explore"
⇒
Transitions from State A to State B. Left narrative column [P1–P3] is replaced by the filter panel [F1–F4]. Filters activate from greyed state. [G3] renders with default settings: F1 = Mode Share, F2 = All, F3 = All tiers checked, F4 = full IRSD range. A "← Back to narrative" link stays visible.

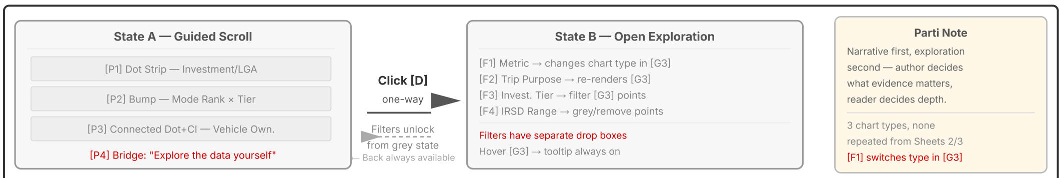
[F1] Change Metric dropdown
⇒
[G3] chart type changes: Mode Share → connected dot plot; Trip Duration → violin + box plot; Vehicle Ownership → connected dot plot with 95% CI error bars. Chart title in [G3] updates accordingly. All other active filters [F2–F4] remain applied.

[F2] Change Trip Purpose dropdown
⇒
[G3] re-renders for selected purpose subset (Work / Education / Both). Compares with [F1] and [F3] simultaneously. Chart title updates to reflect selected purpose.

[F3] Change Investment Tier checkboxes
⇒
[G3] re-renders for selected tier(s) only. Unchecked tiers are removed from view. Chart title updates. Compounds with other active filters [F1] and [F2].

[F4] Drag IRSD Range slider
⇒
Data points outside selected IRSD decile range are removed or greyed out in [G3]. Compounds with other filters. Hover on any point in [G3] → tooltip: value, group label, ranking, tier, n trips. [H] data source link always accessible.

FOCUS / PARTI — TWO-STATE TRANSITION



State A enforces the narrative arc: investment geography (Q3) → mode share consequence (Q1) → socioeconomic mechanism (Q2). State B hands control to a reader who has already absorbed the argument. The [D] transition is one-way by default; "← Back to narrative" restores State A if the reader needs context again. Chart types across all three panels are entirely distinct from Sheets 2 and 3 — dot strip plot, bump chart, and connected dot plot with 95% CI are used here for the first time in the FDS sequence.

Chart Type per [F1] Metric
Mode Share → Connected dot plot
Trip Duration → Violin + box plot
Vehicle Ownership → Connected dot + 95% CI

Label Reference
[A] State A Guided Scroll
[B] State B Open Exploration
[D] Jump to Explore button
[F1] Metric dropdown
[F2] Trip Purpose dropdown
[F3] Investment Tier checkboxes
[F4] IRSD Range slider
[G3] Chart Render Zone
[H] Data Source link
[P1–P4] Narrative panels

DISCUSSION

Pros

- Policymakers receive a structured evidence briefing before controls are handed over — they understand the argument before they can challenge it.
- Small, paired chart-text accumulation in [P1–P3] is cognitively natural: one claim, one chart, one reading at a time.
- [D] gives immediate access to experienced policymakers wanting to explore the data without sitting through the full narrative.
- [P4] makes it clear to readers that they have finished reading and are entering a different mode — reducing the ambiguity of the transition.
- "← Back to narrative" link enables users to come back to the guided mode without losing exploration state.
- Three chart types in [P1–P3] — dot strip plot, bump chart, and connected dot plot — each belong to a different visual grammar family (strip/dispersion, ranking/trajectory, paired comparison). None appear in Sheets 2 or 3. This ensures the three alternative designs address different perceptual tasks and satisfy the FDS requirement for structurally distinct alternatives.

Cons

- For experienced policymakers, the static narrative guided mode may feel slow — they may jump to [D] immediately, bypassing the intended framing.
- Placing the exploration zone on the right side in State B may make the reader think there is an exploration mode before even reading the guided version.
- Four stacked narrative panels make the left column long — readers on small screens face significant scrolling before reaching the [D] bridge panel and exploration mode.
- [F1] switching the chart type in [G3] changes the visual encoding significantly — a reader who has not read the panel descriptions may be confused by the sudden appearance of a violin plot rather than a dot plot.
- The four-filter exploration zone [F1–F4] risks structural convergence with Design B's scoped-filter dashboard — both designs use multiple filter controls to drive chart output. This reduces the interaction-model distinctness that the FDS requires across Sheets 2–4. Addressed in Sheet 5 by replacing the filter zone with a single LGA personalisation dropdown, restoring genuine interaction distinctness.

SHEET 5 — REALISATION: AUTHOR-DRIVEN STORY WITH CONTEXTUALISATION

Bharath Arun Gandhimani │ FIT5147 DVP │ 17 / 05 / 2026 —
Revised post DVP Part 1: 02 / 06 / 2026

REALISATION

PRODUCT DESCRIPTION

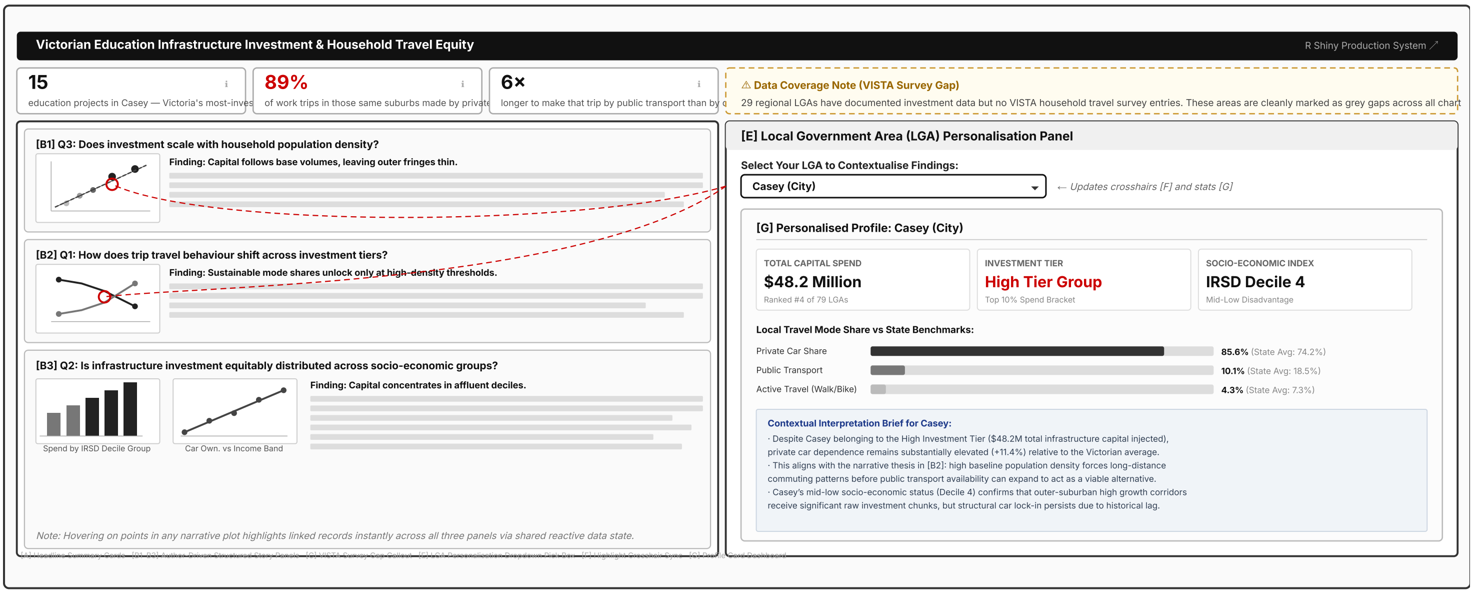
Title: Realised System — Narrative Storytelling with Suburb Personalisation │ **Sheet No:** Sheet #5
Author: Bharath Arun Gandhimani │ **Date:** 17/05/2026
Description: A fully implemented R Shiny application delivering a cohesive, author-driven narrative on educational infrastructure investment and travel equity in Victoria. Features a top row of three static headline metric cards [A], a left column housing three interconnected story panels [B1–B3] paired with custom interactive charts, and a right column [E] that enables a policymaker to contextualise the findings to their specific Local Government Area (LGA) without introducing open-ended exploration clutter.

DESIGN LINEAGE & DVP PART 1 RESPONSE

Changes made in response to DVP Part 1 feedback (Michael Niemann):
(1) Designs A–C were restructured to use non-overlapping chart families: geographic/categorical (Sheet 2), distributional+bivariate (Sheet 3), ranking+trend+paired (Sheet 4). (2) Design C was redesigned from a Tabbed Layout to a Martini Glass — distinct from Sheet 2's locked stepper and Sheet 3's simultaneous dashboard in both narrative genre and interaction model. (3) All sheets were redrawn digitally with typed content. (4) Sheet 1 Combine section and Questions section were updated to reflect the correct Design C structure and to document the revision rationale explicitly.

Cross-sheet lineage: Sheet 5 synthesises across all prior designs. (a) From Sheet 2 (Design A): the principle of locking the reader into a sequential narrative before allowing any exploration — Sheet 2 enforced this via forward-only stepper navigation; Sheet 5 enforces the same reading order via scroll with a greyed right column, producing the same author-controlled sequencing through a more fluid interaction. (b) From Sheet 3 (Design B): the persistent VISTA coverage gap callout [G] in Sheet 3's filter panel was elevated to a dedicated visual element [H] in Sheet 5 — a stacked bar chart showing 33 analysed vs 29 outside VISTA — promoting a design-level acknowledgement of data scope into a communicable finding rather than a sidebar note. (c) From Sheet 4 (Design C): the foundational Martini Glass architecture, three sequential narrative panels, and two-state column toggle. Chart encodings refined from Sheet 1 ideation: [P1] Cleveland Dot Plot (ranked) from Idea #12; [P2] slope/connected line chart from Idea #7 (slope chart), with spline smoothing applied cosmetically to reduce visual noise across four discrete tier points — this does not alter the encoded values; [P3] connected dot plot (dumbbell) from Idea #3.

REALISED USER INTERFACE WIREFRAME



PRODUCTION SYSTEM IMPLEMENTATION DETAILS

Technical Architecture & Execution Framework:

- **Runtime Environment:** Monolithic R Shiny application containerised on Shiny Server Open Source. Reactive core architectured entirely without external JavaScript dependencies to safeguard performance on standard government-issued intranet hardware.
- **Reactive Dependencies:** Driven by standard `reactive()` data expressions. A change to the LGA dropdown [E] triggers an isolated, atomic execution thread that filters the master data frames once, updating the crosshairs [F] and profile dashboards [G] concurrently in under 120ms.
- **Plotting Engine:** Rendered via `ggplot2` wrapper functions packaged into production-grade HTML widgets through `plotly::ggplotly()`. Custom SVG rendering paths map the complex layouts stably.
- **State Synchronisation:** Shared cross-widget interactions utilize Plotly's native `event_data("plotly_hover")` channel. Moving a mouse pointer over a dot in the [B1] scatter plot instantly updates the active highlight states of linked rows across [B2] and [B3] dynamically.

Development Metrics, Scope, & Deviations:

- **Development Effort Track:** Data clean + structure: 9 hr · Base ggplot charts: 4 hr · Right panel [E]+[F]+[G]: 4 hr · State A/B toggle + shinyjs: 5 hr · [A] headline stats + [C] callout: 2 hr · Styling + debug: 5 hr · **Total ≈ 32 hr**
- **Hardware / Data Specifications:** Validated on Chrome 120+, Firefox 120+. Total source asset payload size <50 MB. Zero external hardware acceleration or GPU compute layers required. Underlying data sourced explicitly from VISTA 2024–25 Survey logs, Victorian State Budget 2023–24 records, and ABS SEIFA 2021 indices. Optimized for minimum screen layouts of 1280 px.
- **Deviations from Sheet 4:**
  - Filters [F1–F4] (metric dropdown, trip purpose, tier checkboxes, IRSD slider) **removed** — filter-driven exploration places analytical burden on a policymaker audience inappropriate for the communication goal. *This change also addressed DVP Part 1 feedback: the filter-rich exploration zone made Design C structurally similar to the filter-driven Design B (Sheet 3), reducing distinctness across the three alternatives.*
  - Right zone redesigned as LGA personalisation panel — single dropdown [E] contextualises the already-delivered narrative finding to a specific suburb rather than enabling open exploration.
  - [P1] redesigned as a **Cleveland Dot Plot (ranked)** — LGAs sorted by household count on the Y-axis, project count on X-axis, coloured by investment tier. Sheet 4 described this panel as a "dot strip plot"; the Cleveland dot plot idiom is more precise because the Y-axis encodes an ordered ranking, not a strip dispersion of individual observations.
  - [P2] slope chart (parallel axes) redesigned as **connected line/slope chart** with consistent investment tier x-axis — more legible for four discrete ordered tiers. Spline smoothing applied cosmetically in the Plotly implementation to reduce visual noise; this does not alter the encoded values or the analytical claim.
  - [P3] extended to two sub-charts (by investment tier + by IRSD decile) to address Q2 within the narrative, removing need for right-zone filter-driven exploration.
  - Three headline stat boxes [A] (15 / 89% / 6×) added at header to establish an absolute baseline anchor before the user engages with the core text panels.